

BDX Slow Control Command Quickstart

BDX Slow Control

Document status: Development documentation for the `devel` branch

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Revision date: July 23, 2026

BDX Collaboration

1 Normal daily workflow

This is a command-oriented reference for routine operation. Run the commands from a terminal opened in the Ubuntu graphical desktop or VNC session.

Before operating hardware

Use Phoebus to change PSU or chiller settings. Do not use command-line PV writes unless you have been trained for the specific action. Stopping the software does not switch off hardware outputs or stop the chiller.

1. Enter the repository

```
cd ~/SlowControl/app/bdx-slow-control
```

If an installed command is not found, activate the repository environment in that terminal:

```
source .venv/bin/activate
```

2. Start the complete slow control

```
bdx_slow_control_start overview
```

What it does: starts or reuses the main IOC, checks whether the Raspberry temperature IOC is responding, reports the independent local Archiver Appliance status without changing it, and opens Phoebus on the overview page.

What it does not do: it never starts, stops, audits, repairs, or registers PVs in the Archiver. Use the dedicated `bdx_archiver_*` commands for those actions.

Expected result: the BDX Main IOC terminal remains open when a new IOC is started, and Phoebus displays the BDX overview. The command does not enable PSU outputs, start the chiller, or change setpoints.

Implementation: the installed command is declared in `pyproject.toml` and calls `src/bdx_slow_control/operator_startup.py`; the Phoebus terminal workflow sources `scripts/start_bdx_stack.sh` and launches through the repository Phoebus helpers.

3. Operate from Phoebus

Use the navigation buttons in the overview page.

Page	Use
Overview	Overall status and navigation
PSU	Routine control of LV1 and LV2
Chiller	Chiller status, start/stop, and setpoint
Environment	Raspberry temperature sensors
Trends	Live and archived plots
Global	Global status and confirmed all-off controls

Before changing a setting, check that the relevant communication indicator is healthy and verify the readback after the command.

4. Stop the local software stack

```
bdx_slow_control_kill
```

This gracefully stops the BDX Phoebus session and main IOC. It does not modify the independently managed Archiver Appliance, switch off PSU outputs, send an automatic all-off, stop the chiller, or change hardware settings.

Implementation: `src/bdx_slow_control/operator_commands.py` dispatches this command to `scripts/kill_slow_control_all.sh`.

2 Startup commands

2.1 Open a specific Phoebus page at startup

```
bdx_slow_control_start psu
bdx_slow_control_start chiller
bdx_slow_control_start environment
bdx_slow_control_start trends
```

The same stack is started, but Phoebus opens directly on the selected page.

2.2 Start the Raspberry temperature IOC

```
start-bdx-raspberry-ioc
```

Starts the Raspberry systemd service through SSH and waits for a temperature PV to respond. Run this when the main startup command reports that the Raspberry IOC is unavailable.

2.3 Restart the Raspberry temperature IOC

```
start-bdx-raspberry-ioc --restart
```

Restarts the remote service and verifies that it is responding. The command is implemented by `src/bdx_slow_control/operator_commands.py`; it runs `systemctl start` or `systemctl restart` for `bdx-environment-ioc` on the Raspberry through SSH.

2.4 Reopen Phoebus without starting another IOC

```
./scripts/launch_phoebus.sh overview
```

Use this when the IOC is already running and only the graphical interface must be reopened. Replace `overview` with `psu`, `chiller`, `environment`, `trends`, or `all_pvs`.

3 Read-only checks

```
caproto-get BDX:GLOBAL:SYSTEM_STATE
caproto-get BDX:PSU:LV1:CH1:VOLTAGE_RBV
caproto-get BDX:ENV:TEMP:T00:VALUE
```

These commands read, respectively, the global state, the LV1 channel-1 measured voltage, and the first Raspberry temperature sensor. They do not change hardware settings.

3.1 Check whether a device is reachable

```
nc -vz 172.22.50.20 9221      # LV1
nc -vz 172.22.50.21 9221      # LV2
nc -vz 172.22.50.60 54321     # chiller
```

A successful result means that the TCP control port is reachable. It does not prove that the device is operating correctly.

4 Archiver commands

The Archiver Appliance is managed independently from the main IOC and Phoebus:

Command	Effect	Implementation
<code>bdx_archiver_start</code>	Start and validate the four Archiver components; audit and selectively repair the required catalog unless <code>-no-repair</code> is supplied	<code>src/bdx_slow_control/operator_startup.py</code> , using functions from <code>scripts/start_bdx_stack.sh</code>
<code>bdx_archiver_audit</code>	Check the 18 required archived PVs without changing registrations	<code>src/bdx_slow_control/operator_startup.py</code> , dispatching <code>deploy/archiver-appliance/scripts/repair-archiver.sh</code> with <code>-audit-only</code>
<code>bdx_archiver_repair</code>	Repair only missing, disconnected, or incorrectly configured required PVs	<code>src/bdx_slow_control/operator_startup.py</code> , dispatching <code>deploy/archiver-appliance/scripts/repair-archiver.sh</code>
<code>bdx_archiver_kill</code>	Stop only the Archiver Appliance	<code>src/bdx_slow_control/operator_commands.py</code> , dispatching <code>scripts/kill_slow_control_archiver.sh</code>

Live Channel Access control remains available if historical retrieval is unavailable. Archive history begins only after a PV is successfully registered and sampled; missing earlier data cannot be reconstructed.

5 Shutdown commands

Command	Effect	Script used
<code>bdx_slow_control_kill_phoebus</code>	Stop only BDX slow-control Phoebus processes	<code>scripts/kill_slow_control_phoebus.sh</code>
<code>bdx_archiver_kill</code>	Stop only the repository-managed local Archiver Appliance	<code>scripts/kill_slow_control_archiver.sh</code>
<code>bdx_slow_control_kill_ioc</code>	Stop only the BDX main IOC	<code>scripts/kill_slow_control_ioc.sh</code>
<code>bdx_slow_control_kill</code>	Stop Phoebus and the main IOC without modifying the Archiver Appliance	<code>scripts/kill_slow_control_all.sh</code>

The installed entry points above are declared in `pyproject.toml`; `src/bdx_slow_control/operator_commands.py` validates the repository and forwards their arguments to the listed scripts.

Software shutdown is not hardware shutdown

Put the equipment in the required state before stopping the IOC. None of these commands automatically switches off PSU outputs or stops the chiller.

6 Common problems

Message or symptom	Command or action
<code>command not found</code>	Run <code>source .venv/bin/activate</code> , then retry.
No graphical display is available	Run the startup command from the Ubuntu desktop or VNC session, not from a plain non-graphical SSH shell.
Raspberry IOC is not responding	Run <code>start-bdx-raspberry-ioc</code> ; use <code>start-bdx-raspberry-ioc -restart</code> when a restart is required.
Phoebus was closed but the IOC is running	Run <code>./scripts/launch_phoebus.sh overview</code> .
A PSU is unreachable	Run the corresponding <code>nc -vz</code> check for LV1 or LV2.
The chiller is unreachable	Run <code>nc -vz 172.22.50.60 54321</code> .
Archiver services are absent or degraded at slow-control startup	Start or validate them separately with <code>bdx_archiver_start</code> ; the normal <code>bdx_slow_control_start</code> command intentionally performs only a read-only health report.
Live values are present but history is missing	Run <code>bdx_archiver_audit</code> ; repair with <code>bdx_archiver_repair</code> only after the Archiver components are healthy. Check the selected time range and host clock synchronization.
A value is visible but no longer changes	Check the communication indicator in Phoebus and read the corresponding PV again with <code>caproto-get</code> .

7 Where to find more detail

The *BDX Slow Control User and Developer Manual* contains complete operator procedures, display and PV explanations, architecture, configuration, development, testing, deployment, command implementation references, and release information.